

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	534299
Award Program	2017 New Innovator in Agriculture
Project Title	A Novel Bioassay for Culturing and Characterizing Fastidious Phytopathogens
Grant Start Date	12/05/2017
Grant End Date	12/04/2021
Total Grant Budget	\$599,985.50
Total FFAR Budget	\$299,992.75
Matching Funder(s)	Southern Gardens Citrus
Final Report Date	02/24/2022
Project Director/Principal Investigator Information	
Full Name	Kranthi Mandadi
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Grantee Organization Information	
Organization Name	Texas A&M AgriLife Research
Address	600 John Kimbrough Boulevard, Suite 512
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Authorized Signing Official (ASO) Information	
Name	Julie Bishop
ASO Title	



Email	jbishop@tamu.edu
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General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.
Weslaco, Texas, 34th District of Texas
- 1.2. How many new jobs were created because of this grant funding?
Ten
- 1.3. How many jobs were maintained because of this grant funding?
Ten
- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.
No
- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.
No

2. Scientific Report

- 2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.
Fastidious pathogens cause billions of dollars of agricultural yield losses annually. For example, Candidatus Liberibacter spp. are destructive pathogens, which cause the potato zebra chip and Citrus greening diseases. A major bottleneck in studying these diseases was that the pathogens are unculturable, and there were no high-throughput bioassays to evaluate antimicrobial genes and small molecules. In this FFAR project, we developed an innovative microbial hairy root system for the ex vivo propagation of fastidious pathogens and high throughput efficacy screening of antimicrobials effective against fastidious pathogens. A salient feature of the hairy root system is that it allows four to six times faster screening of antimicrobials, and the platform is versatile. It's useful for the evaluation of disease resistance



genes, antimicrobial peptides, and small molecules.

- 2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-word limit).

The key findings of this project were the development of a high throughput hairy root system that allows ex vivo propagation of fastidious pathogens, and faster (up to 4 times) evaluation of transgene constructs (gain-of-function traits), CRISPR-Cas9 genome editing constructs (loss-of-function traits), and small-molecules therapies effective against fastidious pathogens. The high throughput tools/strategies can be used by the agricultural community to discover new solutions to control devastating plant diseases and enhance plant health.

- 2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as "How did this project lead to improvements in U.S. agriculture and food systems?" or "How can the findings of this study guide or make improvement in future investigations and research?" Outcomes should be explained and classified in one of the following ways:

- a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;
- b) b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and
- c) c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

Potential outcomes: The key findings of the project and high throughput tools/strategies can be used by the agricultural community at large to discover new solutions to control devastating plant diseases and enhance plant health. As such it can be used to spur food system innovation by curtailing crop losses.

Intermediate outcomes: We and others have started leveraging the hairy root system to pursue multiple studies aimed at finding new approaches to control citrus greening and potato zebra chip diseases.

End outcomes: The research impacted the lives of several early-career scientists and students by providing them jobs and educational opportunities to advance their careers in agricultural sciences.



- 2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

Obj. I) Evaluation of the activity of novel antimicrobial peptides in Candidatus Liberibacter solanacearum (CLso)-hairy root cultures for proof-of-concept of gain-of-function studies.

Obj. II) Evaluation of CRISPR genome editing in CLso-hairy root cultures for proof-of-concept of loss-of-function studies.

Obj. III) High-throughput (HTP) screening of small-molecule libraries in CLso-hairy root cultures for proof-of-concept of HTP screening.

Obj. IV) Optimization/proof-of-concept for Candidatus Liberibacter asiaticus (CLas)-citrus hairy root culture production.

- 2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

No

- 2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

All the objectives were accomplished as per the proposed milestones. Results were published in high-impact, peer-reviewed journals such as Nature Communications.

Obj. I) Using genomic and bioinformatics tools, we previously identified several antimicrobial peptides (AMPs) encoded in the plant genomes that could potentially be used to confer tolerance against Candidatus Liberibacter spp. Towards enabling gain-of-function studies, we developed a proof-of-concept evaluation of the antimicrobial activity of eight AMPs using the CLso-potato hairy root system. Briefly, the respective coding sequences were cloned into binary expression vectors and were assayed in the CLso-hairy root cultures. Samples were collected for molecular diagnostics to confirm the expression of defensins and CLso accumulation/titers using RT-qPCR and qPCR techniques, respectively. The results of the antimicrobial assays suggest that five out of eight AMPs confer robust



tolerance against CLso, and also provided proof-of-concept of genetic testing in the hairy root system.

Obj. II) Towards enabling loss-of-function studies, we evaluated CRISPR genome editing in the CLso-potato hairy root system. We developed a proof-of-concept evaluation of CRISPR genome editing feasibility in the CLso-hairy root system towards enabling loss-of-function studies. We also developed new CRISPR-Cas9 vectors amenable for use in the hairy root system. Two approaches were used to test our hypothesis. In the first, we attempted to edit a stably expressing reporter transgene (Green Fluorescent Protein/GFP) in microbial hairy roots. In this experiment, a loss in GFP fluorescence (from the transgene) and the presence of indels at the target GFP site were expected when the roots were transformed with Cas9-sgGFP, but not when transformed with the constructs containing only Cas9 (control). After the experiments, the observed results indicated successful editing of the transgene (GFP) in the CLso hairy root cultures as expected. In a second approach, we evaluated the function of a putative disease susceptibility gene (NPR3) to confer tolerance to CLso. Briefly, multiple sequence alignment and phylogenetic analysis were performed to identify the potato ortholog of Arabidopsis NPR3 (StNPR3) and a CRISPR-Cas9-mediated editing experiment was performed using the CLso-potato hairy root cultures. Constructs containing Cas9 alone (control) or Cas9-sgNPR3 were used to transform healthy or CLso-containing hairy root cultures. Amplicon sequencing of the target site was performed to confirm successful editing of the Cas9-sgNPR3 hairy root cultures, but not in the Cas9 control cultures. At 30 days post-transformation, the CLso levels were estimated for three biological replicates using qPCR. The Cas9-sgNPR3 hairy root cultures had significantly ($p < 0.01$) lower CLso titers than the control, suggesting that NPR3 negatively regulates CLso and providing evidence for the feasibility of evaluating loss-of-function traits in the microbial hairy root system.

Obj. III) To enable high-throughput (HTP) antimicrobial screening of small-molecule libraries, we optimized and developed an in vitro multi-well format assay using the CLso-potato hairy root system. As part of this optimization, we have also evaluated different approaches to isolate high-quality DNA from the hairy root cultures suitable for molecular diagnostics. These tests included evaluation of multiple sample homogenization methods such as using automated bead-beaters and tested commercially available 96-well DNA isolation kits. Importantly, our screening efforts identified ~9 small molecules that showed statistically significant ($p < 0.05$) inhibition of CLso.

Obj. IV) We also expanding the microbial hairy root system concept to the CLas-citrus pathosystem. We successfully generated proof-of-concept for producing CLas-citrus hairy root cultures, albeit with slightly lower efficiency (~40-50%) compared to CLso potato hairy root cultures. We also optimized several factors that could potentially affect the citrus hairy root induction efficiency, and their subsequent propagation such as a) multiple Citrus varieties; b) multiple explant source tissues; c) multiple R. rhizogenes strains with greater virulence and d) environmental growth conditions.



- 2.7. Discuss efforts taken to ensure the approach is scientifically rigorous.

Include approaches taken to ensure robust and unbiased results.

To ensure rigor and reproducibility, all the experiments were performed using recognized scientific/experimental design, methods, and analyses. These include using multiple (three to five) biological replicates. The experiments were repeated twice at independent times. Additionally, the data was independently reviewed by peers and published in high-impact peer-reviewed journals such as Nature Communications.

- 2.8. List key stakeholders that could be served by results of this project.

Academic:

- *Plant pathology scientific community*
- *Early career researchers: postdocs, research scientists*
- *Graduate and undergraduate students*

Industry/Producers:

- *Southern Gardens Citrus, FL*
- *Bayer Crop Science, CA*
- *Citrus Research and Development Foundation, FL*
- *Citrus Research Board, CA*
- *Texas Citrus Mutual, TX*
- *Texas Citrus Pest and Disease management corporation*
- *Wonderful citrus, TX*

- 2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

- *Results were published in high-impact, peer-reviewed journals such as Nature Communications and scientific conferences.*
- *Results were disseminated at multiple meetings with the industry representatives, stakeholders, and partners. During these visits, the stakeholders were updated on some of the results of the project. For instance, how the tools and technologies*



being developed in this project were enabling finding solutions faster to combat fastidious plant pathogens.

- *Additionally, PI was invited to several workshops, panels, conferences, and regional universities, at which time he discussed the impact of the current FFAR project and results.*

2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)
N/A

2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

- *Over the years, the project provided training and professional development for three early-career postdoctoral research associates and research scientists (Ph.D.) and three Hispanic-minority female research associates (M.S.).*
- *The project also trained several Hispanic-minority U.G. student interns in basic concepts of plant genetics and biotechnology, as they contributed to the hairy root projects and transformations.*
- *All the staff and students were mentored by the PI and senior scientists in the lab. To enhance their critical thinking abilities and presentation skills, they were encouraged to present and discuss the project activities in weekly lab meetings, journal clubs, as well as scientific conferences.*

2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

Ten

3. Information Products

3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings,



video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

Scholarly publications

Software

Workshops/panel abstracts

Conference abstracts

Invited seminars/guest lectures

Popular press articles

3.2. Please provide a list of citations for the information products produced during the project.

- Bedre, Renesh H., Avila, C. A., and Mandadi, K. (2021) HTSQualC is a flexible and one-step quality control software for high-throughput sequencing data analysis. *Sci. Rep.* 11, 18725.
- Ramasamy, M., Irigoyen, S., Bedre, R., Avila, C. A., Mandadi, K. K. (2021). Developing zebra chip disease-resistant potato by modulating salicylic acid receptor activity. *American Phytopathological Society (APS) Plant Health (Virtual)*, August 2-6.
- Mora, V., Ramasamy, M., Damaj, M. B., Irigoyen, S., Ancona, V., Ibanez, F., Avila, C., & Mandadi, K. K. (2021). Potato zebra chip: an overview of the disease, control strategies, and prospects. *Front. Microbiol.* 12, 2064.
- Irigoyen, S., Ramasamy, M., Pant, S., Niraula, P., Bedre, R., Gurung, M., Rossi, D., Laughlin, C., Gorman, Z., Achor, D., Levy, A., Kolomiets, M. V., Setamou, M., Badillo-Vargas, I. E., Avila, C. A., Ireby, M., Mandadi, K. K. (2020). Plant hairy roots enable high throughput identification of new antimicrobials against *Candidatus Liberibacter* spp. *Nat. Commun.* 11, 5802.
- Ramasamy, M., Irigoyen, S., Niraula, P., Bedre, R., Pant, S., Avila, C.A., Mandadi, K.K. (2020). Modulation of salicylic acid receptor activity confers resistance to potato zebra chip disease. *American Society of Plant Biologists Worldwide Summit (Virtual)*. 27-31 July
- Mandadi, K. K (2020). Hairy roots for ex vivo propagation of CLas and high-throughput antimicrobial screening. *CLas Culturing Panel/Workshop, Citrus Research and Development Foundation*. November 5.
- Mandadi, K. K (2020) *Biotechnology, and GMOs*, Texas A&M University-Kingsville, Citrus Center, Weslaco, TX. October 22
- Mandadi, K. K (2020) *Genetic and biotechnology strategies to control citrus greening and potato zebra chip diseases*, University of Texas-Rio Grande Valley, Edinburg, TX. February 20.
- Mandadi, K. K., Alabi, O. (2020) *Citrus Greening and Vector-Borne Diseases*, Texas A&M AgriLife Conference Workshop/Panel Discussion, College Station, TX. January 9
- "Hairy roots to the rescue: Speeding up discovery for HLB Management". *Citrograph* 11, 20-22, January 2020.
<http://www.citrusresearch.org/wp-content/uploads/CRB-Citrograph-Mag-Q2-Spring-2020-Web-1.pdf>
- "Joint culturing project for HLB?".



Citrus Industry, Ernie Neff, November 9, 2020.

<https://citrusindustry.net/2020/11/09/joint-culturing-project-for-hlb/>

- "Hairy root matrix used to culture CLas to rapidly screen antimicrobials for improved HLB management".
UCANR Science for Citrus Health. Kranthi Mandadi, Veronica Ancona, Steve Futch, Elizabeth Grafton-Cardwell, Peggy G. Lemaux, Lukasz Stelinski, Monique Rivera & Sara Garcia-Figuera, 2019.
https://ucanr.edu/sites/scienceforcitrushealth/Research_Snapshots/Mandadi/
- Mandadi, K. K (2019) Genomics and biotechnology solutions for agriculture, Texas A&M University High Education Center at McAllen, McAllen, TX. October 30
- Mandadi, K. (2019) Molecular modeling and high throughput antimicrobial screening approaches to identify novel CLas inhibitors. 6th International Research Conference on Huanglongbing, Riverside, CA.
- Mandadi, K. (2019). Novel strategies to detect and control HLB. Texas Citrus Mutual Annual Meeting, McAllen, TX. May 22
- Mandadi, K., Pant S., Laughlin, D., Ancona, V., DaGraca, J. (2019). Updates from the International Research Conference on Huanglongbing (IRCHLB) VI Conference. Texas Citrus Mutual Annual Meeting, McAllen, TX. May 22
- Mandadi, K. K (2019) GMOs and genetic engineering. Plate to the Path Champion Workshop, Weslaco, TX. April 16
- Mandadi, K. (2019). Novel approaches to cultivating and combat fastidious plant pathogens, Seminar for Plant Pathology and Microbiology Dept., Texas A&M University, College Station, TX. February 6
- Mandadi, K. K (2018) Genomics and biotechnology solutions for agriculture, Texas A&M University High Education Center at McAllen, October 22, McAllen, TX.
- Mandadi, K. K. (2018) Development of HLB-tolerant citrus: From gene discovery to commercial deployment. American Society of Horticultural Sciences, July 30-Aug 3, Washington D.C.
- Ramasamy, M. and Mandadi, K. (2018). CRISPR/Cas9 Platforms for Genome Editing in Plants. The Future of Agriculture. TAMU CC student field day, Feb 22-23, Weslaco, TX.

3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

- Bedre, Renesh H., Avila, C. A., and Mandadi, K. (2021) HTSQualC is a flexible and one-step quality control software for high-throughput sequencing data analysis. *Sci. Rep.* 11, 18725.
- Mora, V., Ramasamy, M., Damaj, M. B., Irigoyen, S., Ancona, V., Ibanez, F., Avila, C., & Mandadi, K. K. (2021). Potato zebra chip: an overview of the disease, control strategies, and prospects. *Front. Microbiol.* 12, 2064.
- Irigoyen, S., Ramasamy, M., Pant, S., Niraula, P., Bedre, R., Gurung, M., Rossi, D., Laughlin, C., Gorman, Z., Achor, D., Levy, A., Kolomiets, M. V., Setamou, M., Badillo-Vargas, I. E., Avila, C. A., Irely, M., Mandadi, K. K. (2020). Plant hairy roots enable high throughput identification of new antimicrobials against *Candidatus*

Liberibacter spp. Nat. Commun. 11, 5802.

3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

- "Hairy root matrix used to culture CLas to rapidly screen antimicrobials for improved HLB management".
UCANR Science for Citrus Health. Kranthi Mandadi, Veronica Ancona, Steve Futch, Elizabeth Grafton-Cardwell, Peggy G. Lemaux, Lukasz Stelinski, Monique Rivera & Sara Garcia-Figuera, 2019.
https://ucanr.edu/sites/scienceforcitrushealth/Research_Snapshots/Mandadi/
- "Hairy roots to the rescue: Speeding up discovery for HLB Management".
Citrograph 11, 20-22, January 2020.
<http://www.citrusresearch.org/wp-content/uploads/CRB-Citrograph-Mag-Q2-Spring-2020-Web-1.pdf>

3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	#1	#1	US20210368795A1
Copyrights	#	#	
Trademarks	#	#	
Inventions	#2	#2	

- Mandadi, K. K., Irigoyen, S., Niraula, P., Manikandan, R (2021) Plant antimicrobial compositions and methods of use thereof. US patent application. US20210368795A1
- Mandadi, K.K., Irigoyen, S., Niraula, P., Manikandan, R (2020) Identification of new chemicals and active ingredients to control potato zebra chip and citrus greening diseases. Invention Disclosure 5484AGLR20
- Mandadi, K. K., Irigoyen, S., Pant S., Manikandan, R (2019) A high-throughput gene editing and expression system to evaluate disease susceptibility and resistance genes effective against fastidious pathogens. TAMUS Invention disclosure. 5160AGLR19

3.6. Are any of the information products produced during this grant that are



confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

The microbial hairy root system is currently licensed/optioned to the matching fund's stakeholder, Southern Gardens Citrus, through Texas A&M technology transfer agreements and NDAs.

- 3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that others can access and re-use these data in the future?

The project data and results were published in open-access peer-reviewed scientific journals that can be accessed online free of charge by anyone interested or who would want to use the data. Additionally, we have entertained individual requests for materials or advice and as such extended collaborations to those who are interested in applying strategies we developed.

4. Data Management

- 4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imagery). Please list the data generated for the award.

All the data relating to the project include results that have been reported in open-access peer-reviewed scientific journals and abstracts submitted.

- 4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

N/A

- 4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder. *N/A*



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Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: _____Kranthi Mandadi_____

PI Signature: __________ Date: _____2/24/22_____

Authorized Signing Official (ASO) Name: _____Julie Bishop_____

ASO Signature: __________ Date: _____03/02/2022_____